

UNIFORMLY TIGHT ANCIENT NAVIER–STOKES DYNAMICS AND ENDPOINT L^3 RIGIDITY

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ABSTRACT. We study bounded ancient solutions of the three-dimensional Navier–Stokes equations in backward self-similar variables under uniform L^3 -tightness. We first establish the analytic compactness framework: bounded mild ancient solutions are smooth, locally compact under time translation, and the common $L^\infty_\tau L^3_y$ and tightness bounds used in the compactness statements are stable under locally smooth limits. We then prove a minimal-element reduction for closed normalized uniformly tight collections: any nonempty such collection contains a nonzero solution of minimal $L^\infty(\mathbb{R}^3 \times \mathbb{R})$ size.

For the time-translation hull of a bounded ancient solution we construct compact invariant hulls and invariant probability measures by Krylov–Bogolyubov averaging. These measures satisfy a stationary statistical form of the self-similar Navier–Stokes equation. Passing to the barycenter produces an explicit quadratic covariance term, isolating the obstruction to obtaining a single stationary self-similar profile.

Finally, we close the uniformly L^3 -tight stratum unconditionally. The tail condition, together with the boundedness of the ancient profile, gives the sequence- L^3 bound needed after the physical pullback. Since this pullback preserves the critical L^3 -norm on each time slice, the endpoint ancient Liouville theorem of Albritton–Barker then forces the pullback, and hence the centered profile, to vanish. Therefore the uniformly L^3 -tight stratum of the Type I ancient-profile reduction is excluded without any compact-hull Lyapunov hypothesis.

1. INTRODUCTION AND MAIN RESULTS

1.1. The uniformly tight ancient stratum. We consider the three-dimensional incompressible Navier–Stokes equations in backward self-similar variables. The centered velocity V and pressure P solve

$$(1.1) \quad \partial_\tau V + (V \cdot \nabla)V + \nabla P = \Delta V - \frac{1}{2}(V + y \cdot \nabla V), \quad \operatorname{div} V = 0$$

on $\mathbb{R}^3 \times \mathbb{R}$. In the companion Type I reduction paper [3], admissible Type I singularities are reduced to nonzero normalized ancient solutions of (1.1). The present paper starts after that reduction and studies the uniformly L^3 -tight stratum, meaning the stratum in which

$$(1.2) \quad \lim_{R \rightarrow \infty} \sup_{\tau \in \mathbb{R}} \int_{|y| > R} |V(y, \tau)|^3 dy = 0.$$

Several compactness statements below also record the common bound

$$(1.3) \quad \sup_{\tau \in \mathbb{R}} \|V(\tau)\|_{L^3(\mathbb{R}^3)} < \infty,$$

but the endpoint exclusion of the tight branch uses only boundedness and the tail condition (1.2).

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This profile condition is distinct from the source-level condition

$$u \in L^\infty(0, T; L^3(\mathbb{R}^3)).$$

For finite-energy suitable weak solutions, the source-level assumption $u \in L_t^\infty L_x^3$ near a putative singular time is incompatible with singularity by the theorem of Escauriaza–Seregin–Šverák [4]. Uniform L^3 -tightness in this paper is a condition on centered ancient profiles after blow-up.

1.2. Minimal elements and compact hulls. The first reduction is variational:

$$\text{nonempty tight normalized class} \implies L^\infty\text{-minimal ancient element.}$$

The normalization is a fixed local velocity nonvanishing condition. It prevents minimizing sequences from losing all mass by time translation. Once a minimal element V_* is selected, its time-translation hull is

$$\mathcal{H}(V_*) = \overline{\{\Theta_s V_* : s \in \mathbb{R}\}}^{C_{\text{loc}}^\infty}, \quad (\Theta_s V)(y, \tau) = V(y, \tau + s).$$

The hull is compact because bounded mild ancient solutions are locally smooth with estimates depending only on their common L^∞ bound.

1.3. Invariant measures and covariance. Every compact invariant hull K carries invariant probability measures. Indeed, Krylov–Bogolyubov averages

$$\mu_T = \frac{1}{T} \int_0^T \delta_{\Theta_s W_0} ds$$

have weak subsequential limits. Such invariant measures satisfy a stationary statistical form of (1.1). If

$$\overline{W}(y) = \int_K W(y, 0) d\mu(W),$$

then the averaged nonlinearity decomposes as

$$\int_K W_i W_j d\mu = \overline{W}_i \overline{W}_j + C_{ij}.$$

Thus the barycenter solves the stationary self-similar equation only up to the explicit covariance defect C . The invariant-measure argument alone does not prove that a recurrent compact hull contains a single stationary profile.

1.4. Endpoint L^3 closure of the tight stratum. The compact-hull and invariant-measure constructions identify the covariance obstruction in a purely local topology, but they are not needed to close the uniformly L^3 -tight stratum. The tail condition already gives the sequence- L^3 information required by the endpoint ancient Liouville theorem, because the centered profile is bounded. Under the physical change of variables

$$u(x, t) = (-t)^{-1/2} V\left(\frac{x}{\sqrt{-t}}, -\log(-t)\right), \quad t < 0,$$

the L^3 -norm is invariant:

$$\|u(t)\|_{L_x^3} = \|V(-\log(-t))\|_{L_y^3}.$$

Consequently every tail-tight centered ancient solution produces a physical mild ancient solution with bounded L^3 -norm along a sequence $t_k \downarrow -\infty$. The endpoint theorem of Albritton–Barker [1] then gives $u \equiv 0$, and the change of variables gives $V \equiv 0$.

This is a stratum-local argument. It uses only the boundedness and tail tightness already present in the ancient profile class and introduces no additional estimate outside that stratum, recurrence object, or Lyapunov functional.

The logical status of the paper is summarized as follows.

Step	Status in this paper
bounded ancient smoothing	proved
local smooth compactness	proved
closure of uniform L^3 -tightness	proved
minimal L^∞ element	proved
compact trajectory hull	proved
invariant measures	proved
stationary statistical identity	proved
covariance obstruction	identified
physical pullback and critical L^3 invariance	proved
endpoint L^3 ancient Liouville closure	cited and applied
tight-class input for the companion paper	unconditional corollary

Thus the uniformly L^3 -tight class is closed inside its own stratum. The compact-hull material remains useful for describing possible recurrent dynamics and the covariance obstruction, but no compact-recurrence or Lyapunov assumption enters the tight-class exclusion proved here.

1.5. Main results. The first headline theorem is the analytic compactness and closure statement.

Theorem 1.1 (Compactness and closure of uniformly tight ancient solutions). *Let $\{V_n\}$ be a sequence of bounded mild ancient solutions of (1.1). Assume*

$$\sup_n \|V_n\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} < \infty.$$

Then a subsequence converges locally smoothly to a bounded mild ancient solution V .

If in addition there are $L < \infty$ and a modulus $\omega : [1, \infty) \rightarrow [0, \infty)$, $\omega(R) \rightarrow 0$, such that

$$\sup_n \sup_{\tau \in \mathbb{R}} \|V_n(\tau)\|_{L^3(\mathbb{R}^3)} \leq L$$

and

$$\sup_n \sup_{\tau \in \mathbb{R}} \int_{|y| > R} |V_n(y, \tau)|^3 dy \leq \omega(R) \quad (R \geq 1),$$

then

$$\sup_{\tau \in \mathbb{R}} \|V(\tau)\|_{L^3(\mathbb{R}^3)} \leq L, \quad \sup_{\tau \in \mathbb{R}} \int_{|y| > R} |V(y, \tau)|^3 dy \leq \omega(R),$$

and in particular V is uniformly L^3 -tight.

Compatible pressure representatives also converge locally weakly modulo time-dependent gauges under the additional $L^\infty_\tau L^3_y$ assumption; see [Theorem 3.3](#).

The second headline theorem is the minimal-element reduction.

Theorem 1.2 (Minimal element in a normalized uniformly tight class). *Let A be a nonempty collection of bounded mild ancient solutions satisfying a common L^∞ bound, a common $L^\infty_\tau L^3_y$ bound, a common uniform L^3 -tightness modulus, and a fixed local nonvanishing normalization*

$$\sup_{\tau \in \mathbb{R}} \int_{B_{R_0}} |V(y, \tau)|^3 dy \geq \eta_0 > 0.$$

Assume A is invariant under time translation and closed under locally smooth limits which retain the normalization. Then A contains an element V_ minimizing*

$$\|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})}$$

among all $V \in A$. In particular $V_ \not\equiv 0$.*

The third headline theorem is the endpoint L^3 tight-stratum Liouville criterion.

Theorem 1.3 (Uniformly tight Liouville criterion). *Let A be a collection of bounded mild ancient solutions of (1.1). Assume there is a modulus $\omega : [1, \infty) \rightarrow [0, \infty)$, $\omega(R) \rightarrow 0$, such that*

$$\sup_{\tau \in \mathbb{R}} \int_{|y| > R} |V(y, \tau)|^3 dy \leq \omega(R) \quad \text{for every } V \in A, \ R \geq 1,$$

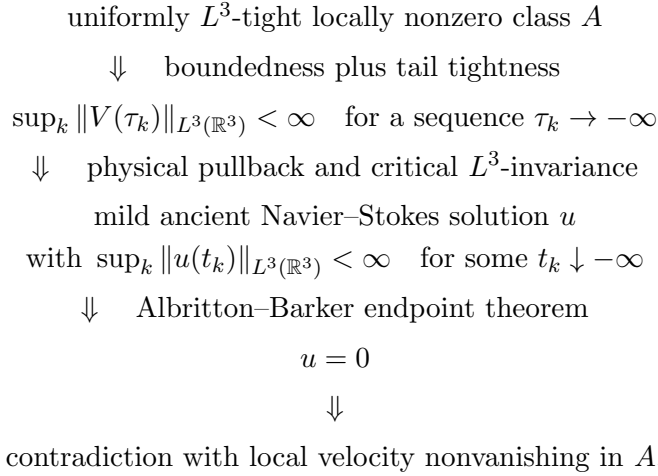
and assume that A has a fixed local nonvanishing normalization:

$$\sup_{\tau \in \mathbb{R}} \int_{B_{R_0}} |V(y, \tau)|^3 dy \geq \eta_0 > 0 \quad \text{for every } V \in A.$$

Then

$$A = \emptyset.$$

The proof of the last theorem follows the dependency diagram



1.6. Relation to the companion paper. The companion paper [3] reduces admissible Type I singularities to normalized Seregin ancient limits and organizes such limits into small, stationary L^3 , uniformly tight, structured/decay, and residual classes. The present paper supplies the tight-class input without an additional compact-dynamical hypothesis: the uniformly L^3 -tight class contains no nonzero normalized ancient limit.

1.7. Organization of the paper. Section 2 fixes the centered equation, mild formulation, time translation, and local smooth topology. Section 3 proves smoothing, pressure reconstruction, stability, and compactness. Section 4 proves closure of uniform L^3 -tightness and recalls stationary L^3 rigidity. Section 5 proves the minimal-element reduction. Section 6 constructs compact trajectory hulls and minimal invariant subsets. Section 7 constructs invariant measures and identifies the covariance obstruction. Section 8 proves the physical pullback identities, applies the endpoint L^3 ancient Liouville theorem, and records the interface with the companion residual-criterion paper.

2. CENTERED ANCIENT SOLUTIONS AND THE LOCAL SMOOTH TOPOLOGY

2.1. The centered equation. Throughout the paper V denotes a centered ancient velocity and P its pressure. A generic trajectory in a compact hull is denoted by W , and a stationary profile by $w(y)$. Define

$$LV = \Delta V - \frac{1}{2}y \cdot \nabla V - \frac{1}{2}V.$$

After applying the Helmholtz–Leray projection $\mathbb{P}$, the centered equation becomes

$$(2.1) \quad \partial_\tau V = LV - \mathbb{P} \operatorname{div}(V \otimes V).$$

2.2. The Ornstein–Uhlenbeck–Stokes semigroup. Let $S(a) = e^{aL}$, $a > 0$. It is given by

$$(2.2) \quad (S(a)f)(y) = e^{-a/2} \int_{\mathbb{R}^3} \frac{\exp\left(-\frac{|z - e^{-a/2}y|^2}{4(1 - e^{-a})}\right)}{[4\pi(1 - e^{-a})]^{3/2}} f(z) dz.$$

The factor $e^{-a/2}$ records the damping caused by the term $-V/2$.

Lemma 2.1 (Centered Stokes kernel estimate). *For $a > 0$ and $\phi \in C_c^\infty(\mathbb{R}^3; \mathbb{R}^3)$,*

$$(2.3) \quad \|(S(a)\mathbb{P} \operatorname{div})^* \phi\|_{L^1(\mathbb{R}^3)} \leq C e^{-a/2} (1 - e^{-a})^{-1/2} \|\phi\|_{L^1(\mathbb{R}^3)}.$$

Consequently the Duhamel term in (2.4) is well-defined by duality for bounded V .

Proof. Set $\lambda = e^{-a/2}$ and $\theta = 1 - e^{-a}$. From (2.2),

$$S(a)f(y) = e^{-a/2} (e^{\theta\Delta} f)(\lambda y).$$

Thus

$$S(a)^* \phi(z) = e^{-a/2} \lambda^{-3} (e^{\theta\Delta} \psi)(z), \quad \psi(x) = \phi(x/\lambda),$$

and $\|\psi\|_{L^1} = \lambda^3 \|\phi\|_{L^1}$. The Leray projection is a homogeneous Fourier multiplier of degree zero, so it commutes with this dilation. The classical heat–Stokes kernel estimate

$$\|(e^{\theta\Delta} \mathbb{P} \operatorname{div})^* \phi\|_{L^1} \leq C \theta^{-1/2} \|\phi\|_{L^1}, \quad \theta > 0,$$

follows from the Oseen kernel bound; see [5, 6, 11]. Substituting $\theta = 1 - e^{-a}$ and using

$$e^{-a/2} \lambda^{-3} \|\psi\|_{L^1} = e^{-a/2} \|\phi\|_{L^1}$$

gives (2.3). Since $(1 - e^{-a})^{-1/2}$ is integrable at $a = 0$, the duality pairing in (2.4) is integrable on finite time intervals for $V \in L^\infty$. $\square$

2.3. Riesz transform convention. We use the convention that R_i has Fourier symbol $i\xi_i/|\xi|$. Thus $R_i R_j$ has symbol $-\xi_i \xi_j / |\xi|^2$, and

$$P = R_i R_j (F_{ij}) \iff -\Delta P = \partial_i \partial_j F_{ij}.$$

Pressures are determined only modulo functions of time unless a whole-space Leray gauge is explicitly chosen.

2.4. Bounded mild ancient solutions.

Definition 2.2 (Bounded mild ancient solution). A vector field $V : \mathbb{R}^3 \times \mathbb{R} \rightarrow \mathbb{R}^3$ is a bounded mild ancient solution of (1.1) if

$$V \in L^\infty(\mathbb{R}^3 \times \mathbb{R}), \quad \operatorname{div} V = 0$$

in distributions, and for every $\sigma < \tau$,

$$(2.4) \quad V(\tau) = S(\tau - \sigma)V(\sigma) - \int_\sigma^\tau S(\tau - s) \mathbb{P} \operatorname{div}(V \otimes V)(s) ds$$

in the weak-* topology of $L^\infty(\mathbb{R}^3)$.

The Duhamel term in (2.4) is interpreted by duality. For $\phi \in C_c^\infty(\mathbb{R}^3; \mathbb{R}^3)$,

$$\langle S(\tau - s) \mathbb{P} \operatorname{div}(V \otimes V)(s), \phi \rangle = \langle V \otimes V(s), (S(\tau - s) \mathbb{P} \operatorname{div})^* \phi \rangle,$$

and Theorem 2.1 makes the scalar integrand absolutely integrable on $s \in (\sigma, \tau)$.

2.5. Time translation. For $s \in \mathbb{R}$ define

$$(\Theta_s V)(y, \tau) = V(y, \tau + s).$$

The equation, the mild formulation, the L^∞ norm, and the $L^\infty_\tau L^3_y$ and tail bounds used below are invariant under Θ_s . Indeed, replacing (σ, τ) by $(\sigma + s, \tau + s)$ in (2.4) and changing variables in the time integral gives the mild formula for $\Theta_s V$.

2.6. The locally smooth topology.

Definition 2.3 (Local smooth topology). The locally smooth topology on $C^\infty_{\text{loc}}(\mathbb{R}^3 \times \mathbb{R}; \mathbb{R}^3)$ is generated by the seminorms

$$\|V\|_{m,N} = \max_{|\alpha|+j \leq m} \sup_{\substack{|y| \leq N \\ |\tau| \leq N}} |\partial_y^\alpha \partial_\tau^j V(y, \tau)|, \quad m, N \in \mathbb{N}.$$

One compatible metric is

$$d(U, V) = \sum_{m,N=1}^{\infty} 2^{-(m+N)} \frac{\|U - V\|_{m,N}}{1 + \|U - V\|_{m,N}}.$$

Thus $V_n \rightarrow V$ locally smoothly if and only if the convergence is in $C^m(K)$ for every compact cylinder $K \Subset \mathbb{R}^3 \times \mathbb{R}$ and every $m \geq 0$.

3. SMOOTHING, PRESSURE RECONSTRUCTION, AND COMPACTNESS

3.1. Local smoothing.

Lemma 3.1 (Smoothing of bounded mild solutions). *Every bounded mild ancient solution is smooth on $\mathbb{R}^3 \times \mathbb{R}$ and satisfies the projected equation (2.1) classically. Moreover, for each compact cylinder $K \Subset \mathbb{R}^3 \times \mathbb{R}$ and each integer $m \geq 0$,*

$$\|V\|_{C^m(K)} \leq C(m, K, \|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})}).$$

The same estimate holds uniformly for any family with a common L^∞ bound.

Proof. Fix a compact time interval $J \Subset \mathbb{R}$ containing the time projection of K , and set $t = -e^{-\tau}$. On the corresponding compact interval $t(J) \Subset (-\infty, 0)$, define

$$U(x, t) = (-t)^{-1/2} V(x/\sqrt{-t}, -\log(-t)).$$

The map $(x, t) \mapsto (x/\sqrt{-t}, -\log(-t))$ is a smooth diffeomorphism between $\mathbb{R}^3 \times t(J)$ and $\mathbb{R}^3 \times J$, with all derivatives bounded on compact subcylinders.

The centered mild formula (2.4) transforms, under this parabolic change of variables, into the usual projected Navier–Stokes mild formula on every compact subinterval of $t(J)$:

$$U(t) = e^{(t-s)\Delta} U(s) - \int_s^t e^{(t-r)\Delta} \mathbb{P} \operatorname{div}(U \otimes U)(r) dr, \quad s < t.$$

This transformation uses only the projected mild equation: the Leray projection is homogeneous of degree zero and commutes with the parabolic dilation, while the Ornstein–Uhlenbeck kernel in (2.2) becomes the heat kernel with time parameter $t - s$. No pressure formulation is used at this stage. The L^∞ norm of U on $\mathbb{R}^3 \times t(J)$ is bounded by a constant depending only on J and $\|V\|_{L^\infty}$.

The standard local smoothing theorem for bounded projected mild Navier–Stokes solutions [7, 6] gives first local derivative bounds away from the initial time in any mild representation. In the present ancient setting one may choose the initial time strictly before the compact subcylinder under consideration. Iterating the differentiated Duhamel formula and then

applying the interior parabolic bootstrap for the Stokes system [9, 11] gives uniform bounds for all derivatives of U on compact subcylinders of $\mathbb{R}^3 \times t(J)$. The mild formula may then be differentiated in time, so U solves the projected Navier–Stokes equation classically. Returning to (y, τ) gives the asserted estimates for V and the classical projected equation (2.1). The argument is uniform for families with a common L^∞ bound. $\square$

3.2. Pressure reconstruction.

Lemma 3.2 (Local pressure reconstruction). *Let V be a smooth bounded solution of the projected centered equation (2.1). Then for every ball B_R and compact interval $I \subseteq \mathbb{R}$ there exists*

$$P_R \in C^\infty(B_R \times I),$$

unique up to an additive function of time, such that

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P_R = \Delta V - \frac{1}{2}(V + y \cdot \nabla V), \quad \operatorname{div} V = 0,$$

on $B_R \times I$.

Proof. Set

$$F = \Delta V - \frac{1}{2}(V + y \cdot \nabla V) - \partial_\tau V - (V \cdot \nabla)V.$$

Since $(V \cdot \nabla)V = \operatorname{div}(V \otimes V)$ and V satisfies (2.1), self-adjointness of the Leray projection gives

$$\int_I \int_{B_R} F \cdot \Phi \, dy \, d\tau = 0$$

for every $\Phi \in C_c^\infty(B_R \times I; \mathbb{R}^3)$ satisfying $\operatorname{div}_y \Phi = 0$. The local de Rham theorem, applied in the spatial variables with τ as a parameter, gives a scalar distribution P_R on $B_R \times I$, unique up to a distribution depending only on time, such that $\nabla_y P_R = F$. Since F is smooth, all spatial derivatives of P_R of positive order are smooth. Fixing, for instance, the spatial mean of P_R over B_R to be zero for each τ selects a representative which is smooth on $B_R \times I$. This proves the local pressure formulation. $\square$

Lemma 3.3 (Leray pressure and local gauges). *Let V be a bounded mild ancient solution satisfying*

$$\sup_{\tau \in \mathbb{R}} \|V(\tau)\|_{L^3(\mathbb{R}^3)} < \infty.$$

Then the whole-space Leray pressure

$$P = R_i R_j (V_i V_j)$$

belongs to $L^\infty(\mathbb{R}; L^{3/2}(\mathbb{R}^3))$. Moreover, if $V_n \rightarrow V$ locally smoothly, the V_n have a common $L_\tau^\infty L_y^3$ bound, and

$$P_n = R_i R_j (V_{n,i} V_{n,j}),$$

then on every compact cylinder $B_R \times [\tau_1, \tau_2]$ there are bounded measurable gauges $a_n(\tau)$ and $a(\tau)$ such that, after passing to a subsequence,

$$P_n - a_n(\tau) \rightharpoonup P - a(\tau) \quad \text{weakly in } L^{3/2}(B_R \times [\tau_1, \tau_2]).$$

Equivalently, after replacing the limiting pressure by an equivalent representative modulo a function of time, the weak limit may be denoted by P .

Proof. The first assertion follows from the Calderon–Zygmund boundedness of Riesz transforms on $L^{3/2}(\mathbb{R}^3)$ [12]:

$$\|P(\tau)\|_{L^{3/2}} \leq C\|V_i(\tau)V_j(\tau)\|_{L^{3/2}} \leq C\|V(\tau)\|_{L^3}^2.$$

For the local stability statement, choose $\chi \in C_c^\infty(B_{4R})$ with $\chi \equiv 1$ on B_{3R} . Decompose on each time slice

$$P_n = R_i R_j (\chi V_{n,i} V_{n,j}) + R_i R_j ((1 - \chi) V_{n,i} V_{n,j}).$$

The local term converges strongly in $L^q(B_{2R} \times [\tau_1, \tau_2])$ for every finite q . This is because $V_n \rightarrow V$ smoothly on $\bar{B}_{4R} \times [\tau_1, \tau_2]$, so $\chi V_{n,i} V_{n,j} \rightarrow \chi V_i V_j$ strongly in every finite L^q , and the Riesz transforms are Calderon–Zygmund operators on L^q , $1 < q < \infty$.

It remains to control the exterior part modulo constants. Let K_{ij} be the convolution kernel of $R_i R_j$. For $x \in B_R$ and $|z| > 4R$,

$$|K_{ij}(x - z) - K_{ij}(-z)| \leq C_R |z|^{-4}.$$

Define the exterior gauges

$$b_n(\tau) = \int_{\mathbb{R}^3} K_{ij}(-z)(1 - \chi(z))V_{n,i}(z, \tau)V_{n,j}(z, \tau) dz$$

and $b(\tau)$ by the same formula with V in place of V_n . The cutoff removes the singularity at the origin, and the integral is absolutely convergent at infinity because, on the dyadic annulus $A_m = \{2^m R < |z| < 2^{m+1} R\}$,

$$\int_{A_m} |z|^{-3} |V_n(z, \tau)|^2 dz \leq C(2^m R)^{-3} \|V_n(\tau)\|_{L^3(A_m)}^2 |A_m|^{1/3} \leq CL^2(2^m R)^{-2}.$$

The same dyadic estimate shows that b_n and b are bounded uniformly on $[\tau_1, \tau_2]$. The maps $\tau \mapsto b_n(\tau)$ and $\tau \mapsto b(\tau)$ are measurable, because they are limits of measurable truncated integrals. After subtracting $b_n(\tau)$, the exterior contribution on B_R is

$$E_n(x, \tau) = \int_{\mathbb{R}^3} [K_{ij}(x - z) - K_{ij}(-z)](1 - \chi(z))V_{n,i}(z, \tau)V_{n,j}(z, \tau) dz.$$

The same dyadic estimate, now using $|z|^{-4}$, gives

$$\|E_n(\cdot, \tau)\|_{L^\infty(B_R)} \leq C(R)L^2$$

uniformly in n and τ . Splitting the exterior integral into $\{4R < |z| < R'\}$ and $\{|z| > R'\}$, the first part converges by local smooth convergence, while the second part is bounded uniformly by CL^2/R' . Letting first $n \rightarrow \infty$ and then $R' \rightarrow \infty$ identifies the local limit of E_n with the corresponding exterior contribution for V . Therefore, with $a_n = b_n$ and $a = b$, the decompositions of $P_n - a_n$ and $P - a$ have local parts converging strongly and exterior parts converging distributionally with a uniform local $L^{3/2}$ bound. Hence

$$P_n - a_n(\tau) \rightharpoonup P - a(\tau) \quad \text{weakly in } L^{3/2}(B_R \times [\tau_1, \tau_2]).$$

□

3.3. Stability under locally smooth limits.

Lemma 3.4 (Closure of the mild formulation). *Let V_n be bounded mild ancient solutions satisfying*

$$\sup_n \|V_n\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} < \infty.$$

If $V_n \rightarrow V$ locally smoothly, then V is a bounded mild ancient solution of (1.1).

Proof. The common L^∞ bound passes to V by pointwise convergence on compact subsets. Divergence freeness passes distributionally. Fix $\sigma < \tau$ and test (2.4) for V_n against $\phi \in C_c^\infty(\mathbb{R}^3; \mathbb{R}^3)$. The endpoint terms are

$$\langle V_n(\tau), \phi \rangle, \quad \langle V_n(\sigma), S(\tau - \sigma)^* \phi \rangle.$$

The first converges by local smooth convergence. For the second, note that $S(\tau - \sigma)^* \phi \in L^1(\mathbb{R}^3)$. Approximate this L^1 function by a compactly supported smooth function and use the common L^∞ bound to control the L^1 approximation error uniformly in n .

For the Duhamel term, write

$$\langle S(\tau - s) \mathbb{P} \operatorname{div}(V_n \otimes V_n)(s), \phi \rangle = \langle V_n(s) \otimes V_n(s), (S(\tau - s) \mathbb{P} \operatorname{div})^* \phi \rangle.$$

For fixed s , the adjoint test belongs to L^1 ; pairing convergence again follows by compact approximation in L^1 , local smooth convergence, and the common L^∞ bound on $V_n \otimes V_n$. The approximation error is uniform in n for this fixed s . Moreover Theorem 2.1 gives the integrable domination

$$C \sup_n \|V_n\|_{L^\infty}^2 e^{-(\tau-s)/2} (1 - e^{-(\tau-s)})^{-1/2} \|\phi\|_{L^1}, \quad s \in (\sigma, \tau).$$

Dominated convergence in s passes the Duhamel integral to the limit. Thus V satisfies (2.4). $\square$

3.4. Compactness of bounded families.

Proposition 3.5 (Compactness of bounded ancient families). *Let $\{V_n\}$ be bounded mild ancient solutions satisfying*

$$\sup_n \|V_n\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} < \infty.$$

Then a subsequence converges locally smoothly to a bounded mild ancient solution V .

Proof. By Theorem 3.1, the sequence is bounded in every C^m -norm on every compact cylinder. Arzela–Ascoli on a countable exhaustion $B_N \times [-N, N]$, followed by a diagonal extraction over (m, N) , gives a subsequence converging locally smoothly to a smooth limit. Theorem 3.4 shows that the limit is again a bounded mild ancient solution. $\square$

4. UNIFORM L^3 -TIGHTNESS AND STATIONARY L^3 RIGIDITY

4.1. Uniform L^3 -tightness.

Definition 4.1 (Uniform L^3 -tightness). A bounded ancient solution V is uniformly L^3 -tight if

$$\lim_{R \rightarrow \infty} \sup_{\tau \in \mathbb{R}} \int_{|y| > R} |V(y, \tau)|^3 dy = 0.$$

For the compactness statements in this section we sometimes assume, in addition, a common $L_\tau^\infty L_y^3$ bound. The endpoint Liouville argument in Section 8 does not take this bound as a separate hypothesis; it obtains the required sequence estimate from boundedness and the tail condition above.

4.2. Closedness under locally smooth convergence.

Lemma 4.2 (Closure of uniform L^3 -tightness). *Let V_n be bounded mild ancient solutions satisfying*

$$\sup_n \sup_{\tau \in \mathbb{R}} \|V_n(\tau)\|_{L^3(\mathbb{R}^3)} \leq L < \infty$$

and assume that they have a common tightness modulus: for some $\omega : [1, \infty) \rightarrow [0, \infty)$ with $\omega(R) \rightarrow 0$,

$$\sup_n \sup_{\tau \in \mathbb{R}} \int_{|y| > R} |V_n(y, \tau)|^3 dy \leq \omega(R), \quad R \geq 1.$$

If $V_n \rightarrow V$ locally smoothly, then

$$\sup_{\tau \in \mathbb{R}} \|V(\tau)\|_{L^3(\mathbb{R}^3)} \leq L$$

and

$$\sup_{\tau \in \mathbb{R}} \int_{|y| > R} |V(y, \tau)|^3 dy \leq \omega(R), \quad R \geq 1.$$

In particular V is uniformly L^3 -tight.

Proof. Fix τ . On B_R , local smooth convergence gives

$$\int_{B_R} |V(y, \tau)|^3 dy = \lim_{n \rightarrow \infty} \int_{B_R} |V_n(y, \tau)|^3 dy \leq L^3.$$

Letting $R \rightarrow \infty$ and using monotone convergence gives $\|V(\tau)\|_{L^3} \leq L$. Since τ was arbitrary, the same bound is uniform in time.

For the tail bound, fix $1 \leq R < R'$. On the bounded annulus $\{R < |y| < R'\}$, dominated convergence gives

$$\int_{R < |y| < R'} |V(y, \tau)|^3 dy = \lim_{n \rightarrow \infty} \int_{R < |y| < R'} |V_n(y, \tau)|^3 dy \leq \omega(R).$$

Letting $R' \rightarrow \infty$ and using monotone convergence gives

$$\int_{|y| > R} |V(y, \tau)|^3 dy \leq \omega(R).$$

The estimate is independent of τ , so taking the supremum in time gives the asserted tail bound. Since $\omega(R) \rightarrow 0$, V is uniformly L^3 -tight. $\square$

Proof of Theorem 1.1. The compactness assertion is Theorem 3.5. The $L^\infty_\tau L^3_y$ and tightness closure assertions follow from Theorem 4.2 with the displayed constants L and ω . $\square$

4.3. Stationary time-translate limits. The stationary self-similar profile equation used below is

$$(4.1) \quad (w \cdot \nabla)w + \nabla p = \Delta w - \frac{1}{2}(w + y \cdot \nabla w), \quad \operatorname{div} w = 0.$$

Theorem 4.3 (Stationary limits under uniform L^3 -tightness). *Let V be uniformly L^3 -tight. Let $\tau_n \in \mathbb{R}$, and suppose*

$$\Theta_{\tau_n} V \rightarrow W$$

locally smoothly. If W is stationary, then $W \equiv 0$.

Proof. The sequence $\Theta_{\tau_n} V$ has the same L^3 bound and tightness modulus as V . By Theorem 4.2, the limit W belongs to $L^\infty_\tau L^3_y$. If W is stationary, write $W(y, \tau) = w(y)$. Then $w \in L^3(\mathbb{R}^3)$.

By Theorem 3.4, W is a bounded mild ancient solution. The smoothness from Theorem 3.1 and the local pressure lemma Theorem 3.2 give smooth local pressures p_N on B_N at time 0. Since $\partial_\tau W = 0$, each p_N satisfies

$$(w \cdot \nabla)w + \nabla p_N = \Delta w - \frac{1}{2}(w + y \cdot \nabla w) \quad \text{on } B_N.$$

On overlaps the gradients of the local pressures agree, so the pressures differ only by constants. Adjusting those constants and passing to the compatible union gives a scalar distribution p on $\mathbb{R}^3$ for which (4.1) holds in distributions. The stationary L^3 rigidity theorem [Theorem 4.4](#) applies and gives $w = 0$. $\square$

4.4. Stationary self-similar rigidity.

Theorem 4.4 (Stationary self-similar rigidity). *Let $w \in C^\infty(\mathbb{R}^3; \mathbb{R}^3) \cap L^3(\mathbb{R}^3)$ solve (4.1) in distributions for some scalar distribution p , with $\operatorname{div} w = 0$. Then $w \equiv 0$.*

Proof. This is the theorem of Nečas, Růžička, and Šverák [10], applied to the stationary Leray self-similar system. The smooth distributional formulation stated here is a special case of their weak $L^3(\mathbb{R}^3)$ theorem. $\square$

5. NORMALIZED COLLECTIONS AND MINIMAL ANCIENT SOLUTIONS

5.1. Closed normalized collections.

Definition 5.1 (Closed normalized collection). A collection A of bounded mild ancient solutions is called a closed normalized collection if:

(i) there is $M < \infty$ such that

$$\|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} \leq M \quad \text{for all } V \in A;$$

(ii) there are $R_0 > 0$ and $\eta_0 > 0$ such that

$$(5.1) \quad \sup_{\tau \in \mathbb{R}} \int_{B_{R_0}} |V(y, \tau)|^3 dy \geq \eta_0 \quad \text{for all } V \in A;$$

(iii) A is invariant under time translation:

$$V \in A, s \in \mathbb{R} \implies \Theta_s V \in A;$$

(iv) if $V_n \in A$, $V_n \rightarrow V$ locally smoothly, and V satisfies the same (R_0, η_0) -nonvanishing condition (5.1), then $V \in A$.

5.2. Time-centering.

Lemma 5.2 (Time-centering). *Let A be a closed normalized collection. If $V \in A$, then for every $\delta > 0$ there exists $s \in \mathbb{R}$ such that*

$$\int_{B_{R_0}} |\Theta_s V(y, 0)|^3 dy \geq \eta_0 - \delta.$$

Proof. By (5.1), the supremum in time of the local L^3 mass is at least η_0 . Choose s within δ of this supremum. Since $(\Theta_s V)(y, 0) = V(y, s)$, the displayed estimate follows. Time-translation invariance keeps $\Theta_s V$ in A . $\square$

5.3. Stability of local nonvanishing.

Lemma 5.3 (Stability of local nonvanishing). *If $V_n \rightarrow V$ locally smoothly and*

$$\int_{B_{R_0}} |V_n(y, 0)|^3 dy \geq \eta_0 - o(1),$$

then

$$\int_{B_{R_0}} |V(y, 0)|^3 dy \geq \eta_0.$$

Proof. Local smooth convergence implies uniform convergence on $\overline{B}_{R_0} \times \{0\}$, hence convergence in $L^3(B_{R_0})$. The map $f \mapsto \int_{B_{R_0}} |f|^3$ is continuous on $L^3(B_{R_0})$, and the claim follows by passing to the limit. $\square$

5.4. Lower semicontinuity of size.

Lemma 5.4 (Lower semicontinuity of L^∞ size). *If $V_n \rightarrow V$ locally smoothly and*

$$\sup_n \|V_n\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} < \infty,$$

then

$$\|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} \leq \liminf_{n \rightarrow \infty} \|V_n\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})}.$$

Proof. Let L_0 be the liminf on the right. If $|V(y_0, \tau_0)| > L_0 + \varepsilon$, choose a subsequence with $\|V_n\|_{L^\infty} \rightarrow L_0$. Local uniform convergence at (y_0, τ_0) gives $|V_n(y_0, \tau_0)| > L_0 + \varepsilon/2$ for all large n , contradicting $\|V_n\|_{L^\infty} \rightarrow L_0$. Hence $|V| \leq L_0$ pointwise. Since the smooth representative has pointwise supremum equal to its essential L^∞ norm, the result follows. $\square$

5.5. Existence of a minimal element.

Theorem 5.5 (Existence of a minimal ancient solution). *Let A be nonempty and closed normalized. Set*

$$m = \inf_{V \in A} \|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})}.$$

Then $0 < m < \infty$, and there exists $V_ \in A$ such that*

$$\|V_*\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} = m.$$

In particular $V_ \not\equiv 0$.*

Proof. The common L^∞ bound gives $m < \infty$. The local nonvanishing gives a positive lower bound:

$$\eta_0 \leq \sup_{\tau \in \mathbb{R}} \int_{B_{R_0}} |V(y, \tau)|^3 dy \leq |B_{R_0}| \|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})}^3,$$

so

$$m \geq (\eta_0/|B_{R_0}|)^{1/3} > 0.$$

Choose $V_n \in A$ with

$$m \leq \|V_n\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} \leq m + \frac{1}{n}.$$

By [Theorem 5.2](#), after replacing V_n by a time translate, which stays in A and preserves the L^∞ norm, we may assume

$$\int_{B_{R_0}} |V_n(y, 0)|^3 dy \geq \eta_0 - \frac{1}{n}.$$

The common L^∞ bound and [Theorem 3.5](#) give a locally smooth subsequential limit V_* , which is a bounded mild ancient solution. By [Theorem 5.3](#),

$$\int_{B_{R_0}} |V_*(y, 0)|^3 dy \geq \eta_0,$$

so V_* retains the fixed (R_0, η_0) -normalization. Closedness of A then gives $V_* \in A$. Finally, [Theorem 5.4](#) gives

$$\|V_*\|_{L^\infty} \leq \liminf_{n \rightarrow \infty} \|V_n\|_{L^\infty} = m.$$

Since m is the infimum over A and $V_* \in A$, equality holds. The positive lower bound on m rules out $V_* \equiv 0$. $\square$

5.6. The uniformly tight normalized class.

Definition 5.6 (Uniformly tight normalized collection). Fix $M, L, R_0, \eta_0 > 0$ and a modulus

$$\omega : [1, \infty) \rightarrow [0, \infty), \quad \omega(R) \rightarrow 0.$$

A collection A of bounded mild ancient solutions is called a uniformly tight normalized collection with data $(M, L, \omega, R_0, \eta_0)$ if:

(1) for every $V \in A$,

$$\|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} \leq M;$$

(2) for every $V \in A$,

$$\sup_{\tau \in \mathbb{R}} \|V(\tau)\|_{L^3(\mathbb{R}^3)} \leq L;$$

(3) for every $V \in A$ and every $R \geq 1$,

$$\sup_{\tau \in \mathbb{R}} \int_{|y| > R} |V(y, \tau)|^3 dy \leq \omega(R);$$

(4) for every $V \in A$,

$$\sup_{\tau \in \mathbb{R}} \int_{B_{R_0}} |V(y, \tau)|^3 dy \geq \eta_0;$$

(5) A is invariant under time translation;

(6) A is closed under locally smooth limits retaining item (4).

Proposition 5.7 (Uniformly tight normalized collections are closed normalized). *Every uniformly tight normalized collection is a closed normalized collection.*

Proof. The L^∞ bound, local nonvanishing, time-translation invariance, and closedness clause in [Theorem 5.1](#) are part of [Theorem 5.6](#). The only non-formal point is that the stated L^3 bound and fixed tightness modulus survive locally smooth limits. This follows from [Theorem 4.2](#). $\square$

Proof of Theorem 1.2. By [Theorem 5.7](#), the collection is closed normalized. [Theorem 5.5](#) gives the asserted nonzero minimizer. $\square$

6. COMPACT TRAJECTORY HULLS

6.1. Trajectory hulls.

Definition 6.1 (Trajectory hull). For a bounded mild ancient solution V , its trajectory hull is

$$\mathcal{H}(V) = \overline{\{\Theta_s V : s \in \mathbb{R}\}}^{C_{\text{loc}}^\infty}.$$

6.2. Compactness of bounded hulls.

Theorem 6.2 (Compact trajectory hull). *If V is a bounded mild ancient solution, then $\mathcal{H}(V)$ is compact in the locally smooth topology and invariant under every time translation Θ_s . Every $W \in \mathcal{H}(V)$ is a bounded mild ancient solution with*

$$\|W\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} \leq \|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})}.$$

Proof. Every translate $\Theta_s V$ is a bounded mild ancient solution with the same L^∞ bound as V . Hence every sequence of translates has a locally smoothly convergent subsequence by [Theorem 3.5](#). If $(W_n) \subset \mathcal{H}(V)$, choose translates $\Theta_{s_n} V$ with $d(W_n, \Theta_{s_n} V) < 1/n$. A subsequence of $\Theta_{s_n} V$ converges locally smoothly, and the corresponding subsequence of W_n has the same limit. Thus $\mathcal{H}(V)$ is sequentially compact. Since the locally smooth topology is metrizable, it is compact.

The L^∞ estimate and the mild equation pass to elements of the hull by local smooth convergence and [Theorem 3.4](#). For invariance, note that Θ_t maps the orbit of V onto itself. If $W_n \rightarrow W$ locally smoothly, then $\Theta_t W_n \rightarrow \Theta_t W$ locally smoothly because time translation only shifts the compact time interval on which the seminorms are evaluated. Hence $\Theta_t \mathcal{H}(V) = \mathcal{H}(V)$. $\square$

Corollary 6.3 (Uniform tightness on hulls). *If V is uniformly L^3 -tight with bound L and modulus ω , then every $W \in \mathcal{H}(V)$ satisfies*

$$\sup_{\tau \in \mathbb{R}} \|W(\tau)\|_{L^3(\mathbb{R}^3)} \leq L$$

and

$$\sup_{\tau \in \mathbb{R}} \int_{|y| > R} |W(y, \tau)|^3 dy \leq \omega(R), \quad R \geq 1.$$

The same estimates hold uniformly on every compact invariant subset $K \subset \mathcal{H}(V)$.

Proof. Every translate of V has the same L^3 bound and tightness modulus. Applying [Theorem 4.2](#) to any sequence of translates converging locally smoothly to $W \in \mathcal{H}(V)$ gives the displayed estimates for W . Taking the supremum over $W \in K \subset \mathcal{H}(V)$ gives the final assertion. $\square$

6.3. Compact invariant sets.

Definition 6.4 (Compact invariant set). A nonempty compact subset K of the locally smooth space of bounded mild ancient solutions is invariant if

$$\Theta_t K = K \quad \text{for every } t \in \mathbb{R}.$$

It is minimal if it contains no proper nonempty compact invariant subset.

Lemma 6.5 (Continuity of the translation action). *Let K be a compact invariant set. For each $t \in \mathbb{R}$, $\Theta_t : K \rightarrow K$ is a homeomorphism, and the action*

$$\mathbb{R} \times K \ni (t, W) \mapsto \Theta_t W \in K$$

is jointly continuous.

Proof. For fixed t , convergence $W_n \rightarrow W$ locally smoothly implies

$$\|\Theta_t W_n - \Theta_t W\|_{C^m(B_R \times [-T, T])} = \|W_n - W\|_{C^m(B_R \times [-T+t, T+t])} \rightarrow 0.$$

Thus Θ_t is continuous, and its inverse is Θ_{-t} . If $t_n \rightarrow t$ and $W_n \rightarrow W$, then on any fixed compact cylinder the difference

$$\Theta_{t_n} W_n - \Theta_t W$$

is bounded by a term involving $W_n - W$ on a slightly larger compact cylinder and a term involving the uniform continuity of finitely many derivatives of W on that larger cylinder. Both terms tend to zero, proving joint continuity. $\square$

6.4. Minimal compact invariant subsets.

Proposition 6.6 (Minimal compact invariant subsets). *Let K be a nonempty compact invariant set. Then K contains a nonempty compact minimal invariant subset.*

Proof. Let $\mathcal{F}$ be the family of nonempty compact invariant subsets of K , ordered by reverse inclusion. For a chain $\{K_\alpha\}$, the intersection $\bigcap_\alpha K_\alpha$ is nonempty by compactness and the finite intersection property, since the chain is totally ordered by ordinary inclusion. The intersection is compact and invariant: invariance follows from $\Theta_t K_\alpha = K_\alpha$ for all α and the fact that Θ_t is a homeomorphism. Thus every chain has an upper bound in the reverse-inclusion order. Zorn's lemma gives a maximal element for that order, equivalently a minimal compact invariant subset under ordinary inclusion. $\square$

6.5. The zero trajectory.

Lemma 6.7 (The zero trajectory). *The singleton $\{0\}$ is compact and invariant. If a minimal compact invariant set contains the zero trajectory, then it equals $\{0\}$.*

Proof. The zero solution satisfies $\Theta_t 0 = 0$ for all t . Therefore $\{0\}$ is a nonempty compact invariant subset of any invariant set containing zero. Minimality forces equality. $\square$

7. INVARIANT MEASURES AND THE COVARIANCE OBSTRUCTION

7.1. Krylov–Bogolyubov averages. The following is the standard Krylov–Bogolyubov averaging argument [8, 2].

Theorem 7.1 (Krylov–Bogolyubov averages). *Let K be a compact invariant set and $W_0 \in K$. For $T > 0$, define*

$$\mu_T = \frac{1}{T} \int_0^T \delta_{\Theta_s W_0} ds.$$

Then every sequence $T_n \rightarrow \infty$ has a subsequence such that

$$\mu_{T_n} \rightharpoonup \mu$$

weakly as Borel probability measures on K , and every such limit μ is invariant:

$$(\Theta_t)_\# \mu = \mu \quad \text{for all } t \in \mathbb{R}.$$

If K is minimal, then every invariant Borel probability measure on K has support equal to K .

Proof. The orbit map $s \mapsto \Theta_s W_0$ is continuous by Theorem 6.5; thus μ_T is the push-forward of normalized Lebesgue measure on $[0, T]$. Since K is compact and metrizable, the space of Borel probability measures on K is weakly sequentially compact [2]. Hence μ_{T_n} has a weakly convergent subsequence.

For $F \in C(K)$ and $t > 0$,

$$\int_K F(\Theta_t W) d\mu_T(W) - \int_K F(W) d\mu_T(W) = \frac{1}{T} \int_T^{T+t} F(\Theta_s W_0) ds - \frac{1}{T} \int_0^t F(\Theta_s W_0) ds.$$

The absolute value is at most $2t\|F\|_{C(K)}/T$. Letting $T = T_{n_k} \rightarrow \infty$ gives invariance for $t > 0$; negative times follow by applying the positive-time result to $F \circ \Theta_t$.

If K is minimal and μ is invariant, then $\text{supp } \mu$ is nonempty, closed, and invariant. Indeed, if U is a neighborhood of $\Theta_t W$, then $\Theta_{-t}U$ is a neighborhood of W , and invariance shows $\mu(U) = \mu(\Theta_{-t}U) > 0$ whenever $W \in \text{supp } \mu$. Hence $\Theta_t(\text{supp } \mu) \subset \text{supp } \mu$. Applying the same argument with $-t$ gives the reverse inclusion, so $\text{supp } \mu$ is invariant. It is nonempty because $\mu(K) = 1$, and it is compact because it is closed in compact K . Minimality forces $\text{supp } \mu = K$. $\square$

7.2. Support on minimal hulls.

Corollary 7.2 (Nonzero support). *Let K be a minimal compact invariant set with $K \neq \{0\}$. Then zero is not in K , and every invariant probability measure on K has support equal to K .*

Proof. If zero were in K , [Theorem 6.7](#) would give $K = \{0\}$, contrary to the assumption. The support conclusion is the minimal-support part of [Theorem 7.1](#). $\square$

7.3. Stationary statistical identity.

Theorem 7.3 (Stationary statistical form). *Let K be a compact invariant set and μ an invariant Borel probability measure on K . Then for every solenoidal $\phi \in C_c^\infty(\mathbb{R}^3; \mathbb{R}^3)$,*

$$(7.1) \quad 0 = \int_K \left[\int_{\mathbb{R}^3} W(y, 0) \cdot \left(\Delta \phi + \phi + \frac{1}{2} y \cdot \nabla \phi \right) dy + \int_{\mathbb{R}^3} W_i(y, 0) W_j(y, 0) \partial_j \phi_i(y) dy \right] d\mu(W).$$

Proof. Define

$$F(W) = \int_{\mathbb{R}^3} W(y, 0) \cdot \phi(y) dy.$$

Local smooth convergence makes F continuous on K : if $W_n \rightarrow W$, then $W_n(\cdot, 0) \rightarrow W(\cdot, 0)$ uniformly on the compact support of ϕ . Invariance gives

$$\int_K \frac{F(\Theta_h W) - F(W)}{h} d\mu(W) = 0 \quad (0 < |h| \leq 1).$$

Choose R with $\text{supp } \phi \subset B_R$. On $\overline{B}_R \times [-1, 1]$, compactness of K in the locally smooth topology gives

$$M_\phi = \sup_{W \in K} \|\partial_\tau W\|_{L^\infty(\overline{B}_R \times [-1, 1])} < \infty.$$

For $0 < |h| \leq 1$,

$$\frac{F(\Theta_h W) - F(W)}{h} = \int_{\mathbb{R}^3} \left(\frac{1}{h} \int_0^h \partial_\tau W(y, s) ds \right) \cdot \phi(y) dy,$$

and the absolute value is bounded by $|B_R| \|\phi\|_{L^\infty} M_\phi$, uniformly in W and h . For each fixed W , smoothness gives uniform convergence of the inner average to $\partial_\tau W(y, 0)$ on $\text{supp } \phi$. Dominated convergence on K yields

$$\int_K \int_{\mathbb{R}^3} \partial_\tau W(y, 0) \cdot \phi(y) dy d\mu(W) = 0.$$

For each W , the smooth local equation [\(1.1\)](#) at time zero, tested against solenoidal ϕ , gives

$$\begin{aligned} \int_{\mathbb{R}^3} \partial_\tau W \cdot \phi dy &= \int_{\mathbb{R}^3} W \cdot \Delta \phi dy + \int_{\mathbb{R}^3} W \cdot \phi dy + \frac{1}{2} \int_{\mathbb{R}^3} W \cdot (y \cdot \nabla \phi) dy \\ &\quad + \int_{\mathbb{R}^3} W_i W_j \partial_j \phi_i dy. \end{aligned}$$

The pressure term vanishes because $\text{div } \phi = 0$; the drift identity uses

$$\int y_j \partial_j W_i \phi_i dy = -3 \int W \cdot \phi dy - \int W \cdot (y \cdot \nabla \phi) dy.$$

The right-hand side is a continuous bounded function of $W \in K$. Averaging over K gives [\(7.1\)](#). $\square$

7.4. Barycenter and covariance.

Theorem 7.4 (Barycenter and covariance). *Let K and μ be as in [Theorem 7.3](#). Define*

$$\overline{W}(y) = \int_K W(y, 0) d\mu(W)$$

and

$$C_{ij}(y) = \int_K W_i(y, 0)W_j(y, 0) d\mu(W) - \overline{W}_i(y)\overline{W}_j(y).$$

Then $\overline{W}$ and C are C_{loc}^∞ , $\overline{W}$ is solenoidal, and for every solenoidal $\phi \in C_c^\infty(\mathbb{R}^3; \mathbb{R}^3)$,

$$(7.2) \quad \int_{\mathbb{R}^3} \overline{W} \cdot \left(\Delta \phi + \phi + \frac{1}{2} y \cdot \nabla \phi \right) dy + \int_{\mathbb{R}^3} \overline{W}_i \overline{W}_j \partial_j \phi_i dy + \int_{\mathbb{R}^3} C_{ij} \partial_j \phi_i dy = 0.$$

In particular, $\overline{W}$ is a weak stationary self-similar solution against solenoidal test fields if and only if

$$\int_{\mathbb{R}^3} C_{ij} \partial_j \phi_i dy = 0$$

for every solenoidal test field ϕ .

Proof. Compactness of K in the locally smooth topology gives, for every compact ball and every derivative order, a uniform bound on $\partial_y^\alpha W(y, 0)$ for $W \in K$. Hence one may differentiate $\overline{W}$ under the integral sign. More explicitly, for every multiindex α and every y in a fixed compact ball,

$$\partial_y^\alpha \overline{W}(y) = \int_K \partial_y^\alpha W(y, 0) d\mu(W),$$

because the difference quotients are dominated by the next derivative seminorm, uniformly on K . The same argument applies to the second moment

$$M_{ij}(y) = \int_K W_i(y, 0)W_j(y, 0) d\mu(W),$$

using Leibniz' rule and the uniform bounds on derivatives of W . This gives C_{loc}^∞ regularity of $\overline{W}$, M , and $C = M - \overline{W} \otimes \overline{W}$. Since every W is divergence free, Fubini against scalar test functions gives $\text{div } \overline{W} = 0$: for $\eta \in C_c^\infty(\mathbb{R}^3)$,

$$\int_{\mathbb{R}^3} \overline{W} \cdot \nabla \eta dy = \int_K \int_{\mathbb{R}^3} W(y, 0) \cdot \nabla \eta(y) dy d\mu(W) = 0.$$

For each compactly supported test field ϕ , the maps

$$W \mapsto \int_{\mathbb{R}^3} W(y, 0) \cdot \phi(y) dy, \quad W \mapsto \int_{\mathbb{R}^3} W_i(y, 0)W_j(y, 0) \partial_j \phi_i(y) dy$$

are continuous and bounded on K , again by compactness in C_{loc}^∞ . Thus all integrals below are finite, and Fubini's theorem applies.

Applying [Theorem 7.3](#) and passing the linear term through the K -integral gives the first integral in (7.2). Passing the nonlinear term through the integral gives

$$\int_K W_i W_j d\mu = \overline{W}_i \overline{W}_j + C_{ij}.$$

Substitution yields (7.2). Comparing this identity with the pressure-free stationary weak formulation for $\overline{W}$ gives the stated equivalence with vanishing covariance contribution. $\square$

8. ENDPOINT L^3 CLOSURE OF THE UNIFORMLY TIGHT STRATUM

8.1. Physical pullback. Let V be a smooth centered ancient velocity and let P be a compatible local pressure. For $t < 0$ define

$$(8.1) \quad u(x, t) = (-t)^{-1/2} V\left(\frac{x}{\sqrt{-t}}, -\log(-t)\right),$$

and

$$(8.2) \quad p(x, t) = (-t)^{-1} P\left(\frac{x}{\sqrt{-t}}, -\log(-t)\right).$$

Equivalently,

$$V(y, \tau) = e^{-\tau/2} u(e^{-\tau/2} y, -e^{-\tau}), \quad P(y, \tau) = e^{-\tau} p(e^{-\tau/2} y, -e^{-\tau}).$$

Lemma 8.1 (Pullback solves physical Navier–Stokes). *Let V be a bounded mild ancient solution of (1.1). Then the pullback u in (8.1) is a smooth divergence-free solution of*

$$(8.3) \quad \partial_t u + (u \cdot \nabla) u + \nabla p = \Delta u, \quad \operatorname{div} u = 0,$$

on $\mathbb{R}^3 \times (-\infty, 0)$, with pressure p locally given by (8.2). Moreover, for every $T < 0$,

$$\|u\|_{L^\infty(\mathbb{R}^3 \times (-\infty, T])} \leq (-T)^{-1/2} \|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})}.$$

Proof. By Theorem 3.1, V is smooth. By Theorem 3.2, on every compact parabolic cylinder there is a smooth pressure representative P , unique up to a function of τ , for which (1.1) holds classically. Adding a function of τ to P only adds a function of t to p , and hence does not affect ∇p .

Set

$$\lambda(t) = (-t)^{-1/2}, \quad y = \lambda(t)x, \quad \tau = -\log(-t).$$

Then $\dot{\tau} = \lambda^2$ and

$$\partial_t y = \frac{1}{2} \lambda^2 y, \quad \dot{\lambda} = \frac{1}{2} \lambda^3.$$

Using $u = \lambda V(y, \tau)$, one obtains

$$\partial_t u = \lambda^3 \left(\partial_\tau V + \frac{1}{2} y \cdot \nabla V + \frac{1}{2} V \right),$$

while

$$(u \cdot \nabla_x) u = \lambda^3 (V \cdot \nabla_y) V, \quad \Delta_x u = \lambda^3 \Delta_y V, \quad \nabla_x p = \lambda^3 \nabla_y P.$$

Substituting these identities into (1.1) gives (8.3). The divergence identity follows similarly:

$$\operatorname{div}_x u = \lambda^2 \operatorname{div}_y V = 0.$$

Finally, if $t \leq T < 0$, then $(-t)^{-1/2} \leq (-T)^{-1/2}$, and the stated L^∞ bound follows directly from (8.1). $\square$

8.2. Critical L^3 invariance.

Lemma 8.2 (Invariance of the critical norm). *Let u be the physical pullback of V . Then, for every $t < 0$,*

$$(8.4) \quad \|u(t)\|_{L^3(\mathbb{R}^3)} = \|V(-\log(-t))\|_{L^3(\mathbb{R}^3)}.$$

Proof. With $y = x/\sqrt{-t}$, (8.1) gives

$$|u(x, t)|^3 = (-t)^{-3/2} |V(y, -\log(-t))|^3, \quad dx = (-t)^{3/2} dy.$$

Multiplying these two identities and integrating over $\mathbb{R}^3$ gives (8.4). $\square$

Lemma 8.3 (Tail tightness gives an endpoint sequence). *Let V be a bounded function on $\mathbb{R}^3 \times \mathbb{R}$ satisfying*

$$\lim_{R \rightarrow \infty} \sup_{\tau \in \mathbb{R}} \int_{|y| > R} |V(y, \tau)|^3 dy = 0.$$

Then, for every sequence $\tau_k \in \mathbb{R}$,

$$\sup_k \|V(\tau_k)\|_{L^3(\mathbb{R}^3)} < \infty.$$

Proof. Let $M = \|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})}$. Choose $R \geq 1$ such that

$$\sup_{\tau \in \mathbb{R}} \int_{|y| > R} |V(y, \tau)|^3 dy \leq 1.$$

Then, for every k ,

$$\int_{\mathbb{R}^3} |V(y, \tau_k)|^3 dy \leq M^3 |B_R| + \sup_{\tau \in \mathbb{R}} \int_{|y| > R} |V(y, \tau)|^3 dy \leq M^3 |B_R| + 1.$$

This gives the claimed endpoint sequence bound. $\square$

8.3. Mildness after finite terminal-time restriction.

Lemma 8.4 (Physical pullbacks are mild ancient solutions on finite intervals). *Let V be a bounded mild ancient solution of (1.1), and let u be its physical pullback. For every $T < 0$, the restriction of u to $\mathbb{R}^3 \times (-\infty, T]$ is a bounded mild ancient solution of the physical Navier–Stokes equations after translating the terminal time T to 0.*

Proof. By Theorem 8.1, u is a smooth bounded classical solution on $\mathbb{R}^3 \times (-\infty, T]$. Let $s < t \leq T$. The classical equation on the finite slab $[s, t]$ gives, for every solenoidal $\phi \in C_c^\infty(\mathbb{R}^3; \mathbb{R}^3)$, the weak Duhamel identity

$$\begin{aligned} \langle u(t), \phi \rangle &= \langle e^{(t-s)\Delta} u(s), \phi \rangle \\ &\quad - \int_s^t \left\langle e^{(t-r)\Delta} \mathbb{P} \operatorname{div}(u \otimes u)(r), \phi \right\rangle dr. \end{aligned}$$

Equivalently,

$$u(t) = e^{(t-s)\Delta} u(s) - \int_s^t e^{(t-r)\Delta} \mathbb{P} \operatorname{div}(u \otimes u)(r) dr$$

in the weak-* topology of $L^\infty(\mathbb{R}^3)$. The heat–Stokes kernel bound used in Theorem 2.1 makes the integral finite on every finite interval because u is bounded on $\mathbb{R}^3 \times (-\infty, T]$. Thus the restriction is a bounded mild ancient solution. The time translate $u_T(x, \sigma) = u(x, \sigma + T)$, $\sigma \leq 0$, has the same property on $(-\infty, 0]$. $\square$

8.4. Endpoint L^3 ancient Liouville theorem.

Theorem 8.5 (Albritton–Barker endpoint ancient Liouville theorem). *Let u be a bounded mild ancient solution of the three-dimensional Navier–Stokes equations on $\mathbb{R}^3 \times (-\infty, 0]$. If there exists a sequence $s_k \downarrow -\infty$ such that*

$$\sup_k \|u(s_k)\|_{L^3(\mathbb{R}^3)} < \infty,$$

then $u \equiv 0$.

Proof. This is the endpoint ancient Liouville theorem of Albritton–Barker [1]. The formulation above is the translated terminal-time version needed here. $\square$

Theorem 8.6 (Tail-tight centered ancient solutions vanish). *Let V be a bounded mild ancient solution of (1.1). Assume*

$$\lim_{R \rightarrow \infty} \sup_{\tau \in \mathbb{R}} \int_{|y| > R} |V(y, \tau)|^3 dy = 0.$$

Then $V \equiv 0$.

Proof. Let u be the physical pullback of V . Fix $T < 0$. By Theorem 8.4, after translating the terminal time T to 0, the restriction of u to $(-\infty, T]$ is a bounded mild ancient solution. Choose $t_k = -e^k$. For all sufficiently large k , $t_k < T$, and $\tau_k = -\log(-t_k) = -k$. By Theorem 8.3,

$$\sup_k \|V(-k)\|_{L^3(\mathbb{R}^3)} < \infty.$$

Hence (8.4) gives

$$\|u(t_k)\|_{L^3(\mathbb{R}^3)} = \|V(-k)\|_{L^3(\mathbb{R}^3)}$$

uniformly along the same sequence, after discarding the finitely many indices for which $t_k \geq T$. Set $u_T(x, \sigma) = u(x, \sigma + T)$ for $\sigma \leq 0$ and $s_k = t_k - T$. Then $s_k \downarrow -\infty$ and

$$\|u_T(s_k)\|_{L^3(\mathbb{R}^3)} = \|u(t_k)\|_{L^3(\mathbb{R}^3)}$$

is uniformly bounded. Applying Theorem 8.5 to u_T gives $u_T \equiv 0$ on $\mathbb{R}^3 \times (-\infty, 0]$, equivalently $u = 0$ on $\mathbb{R}^3 \times (-\infty, T]$. Since $T < 0$ was arbitrary, $u = 0$ on $\mathbb{R}^3 \times (-\infty, 0)$. The inverse change of variables then gives

$$V(y, \tau) = e^{-\tau/2} u(e^{-\tau/2} y, -e^{-\tau}) = 0$$

for every (y, τ) . □

9. TIGHT-STRATUM LIOUVILLE CRITERION AND TYPE I INTERFACE

9.1. Normalized tight collections.

Theorem 9.1 (No nonempty normalized uniformly tight collection). *No uniformly tight normalized collection in the sense of Theorem 5.6 is nonempty.*

Proof. Suppose A is such a collection and choose $V \in A$. By the tightness modulus in item (3) of Theorem 5.6, V satisfies the tail condition in Theorem 8.6. Hence Theorem 8.6 gives $V \equiv 0$. This contradicts the local velocity normalization in the same definition:

$$\sup_{\tau \in \mathbb{R}} \int_{B_{R_0}} |V(y, \tau)|^3 dy \geq \eta_0 > 0.$$

Hence A is empty. □

Proof of Theorem 1.3. If A were nonempty, choose $V \in A$. The common tail modulus gives the tail condition in Theorem 8.6, so $V \equiv 0$. This contradicts the fixed local nonvanishing normalization in Theorem 1.3. □

9.2. Interface with the companion residual criterion.

Definition 9.2 (Tight-class Liouville input). The tight-class Liouville input is the assertion that no normalized Seregina ancient limit generated by an admissible Type I source and satisfying the Paper I compact velocity nonvanishing normalization is uniformly L^3 -tight.

Lemma 9.3 (From Paper I velocity nonvanishing to local velocity normalization). *Let V be a smooth bounded normalized Seregin ancient limit. Suppose that the Paper I extraction gives a compact velocity lower bound: there are a compact set $K \subseteq \mathbb{R}^3 \times \mathbb{R}$ and $\eta_I > 0$ such that*

$$\iint_K |V(y, \tau)|^3 dy d\tau \geq \eta_I.$$

Then there exist $R_0 > 0$ and $\eta_0 > 0$ such that

$$\sup_{\tau \in \mathbb{R}} \int_{B_{R_0}} |V(y, \tau)|^3 dy \geq \eta_0.$$

Proof. Choose $R_0 > 0$ and a compact interval $I = [a, b]$, $b > a$, such that $K \subset B_{R_0} \times I$. Then

$$\eta_I \leq \int_a^b \int_{B_{R_0}} |V(y, \tau)|^3 dy d\tau \leq (b - a) \sup_{\tau \in \mathbb{R}} \int_{B_{R_0}} |V(y, \tau)|^3 dy.$$

Thus the conclusion holds with $\eta_0 = \eta_I / (b - a)$. $\square$

Corollary 9.4 (Tight-class input for the residual criterion). *Theorem 9.2 holds.*

Proof. Suppose, contrary to Theorem 9.2, that the companion residual-criterion paper produces a normalized Seregin ancient limit V in the uniformly L^3 -tight class from an admissible Type I source. By the Paper I extraction theorem, V carries compact velocity nonvanishing: there are $K \subseteq \mathbb{R}^3 \times \mathbb{R}$ and $\eta_I > 0$ with

$$\iint_K |V(y, \tau)|^3 dy d\tau \geq \eta_I.$$

In particular $V \not\equiv 0$. Since V is assumed uniformly L^3 -tight, it satisfies the tail condition

$$\lim_{R \rightarrow \infty} \sup_{\tau \in \mathbb{R}} \int_{|y| > R} |V(y, \tau)|^3 dy = 0.$$

Theorem 8.6 then gives $V \equiv 0$, contradicting the compact velocity lower bound above. Hence no normalized Seregin ancient limit generated by an admissible Type I source can belong to the uniformly L^3 -tight class. $\square$

The tight-stratum input used in the companion residual criterion is therefore unconditional. The proof uses only boundedness, the tail-tightness component of the uniformly L^3 -tight stratum, the local-in-time physical pullback, and the endpoint ancient Liouville theorem of Albritton–Barker.

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